

ROLLESTON, Australia

WMO: 943960

Lat: 24.4615S

Lon: 148.6260E

Elev: 722

StdP: 14.32

Time Zone: 10.00 (E10)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 37.7	(c) 40.4	(d) 18.0	(e) 14.0	(f) 65.0	(g) 21.8	(h) 16.8	(i) 61.5	(j) 15.6	(k) 61.1	(l) 14.0	(m) 63.0	(n) 3.0	(o) 150	(p) 0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 21.1	(c) 100.3	(d) 69.7	(e) 97.8	(f) 69.3	(g) 95.2	(h) 69.0	(i) 76.1	(j) 86.1	(k) 74.9	(l) 84.7	(m) 73.9	(n) 84.1	(o) 6.9	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	129.7	79.4	72.5	123.6	78.6	71.2	118.4	77.7	40.1	84.8	38.9	84.5	38.0	84.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.6	(b) 13.9	(c) 12.3	(e) 33.0	(f) 104.7	(g) 1.5	(h) 3.7	(i) 31.9	(j) 107.4	(k) 31.0	(l) 109.6	(m) 30.1	(n) 111.7	(o) 29.0	(p) 114.4		
(4) 15.6	(5) 13.9	(6) 12.3	(7) 33.0	(8) 104.7	(9) 1.5	(10) 3.7	(11) 31.9	(12) 107.4	(13) 31.0	(14) 109.6	(15) 30.1	(16) 111.7	(17) 29.0	(18) 114.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.1	82.1	80.4	77.9	71.5	64.5	59.4	58.7	60.9	67.1	72.8	77.9	80.8		
	DBStd	9.65	4.01	4.52	4.01	3.86	5.01	4.78	5.00	4.79	5.46	5.18	4.82	4.57		
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0		
	HDD65	641	0	0	0	5	70	174	203	148	36	5	0	0		
	CDD50	7709	995	852	865	644	449	283	271	336	514	707	838	955		
	CDD65	2874	530	432	400	199	54	7	8	19	99	248	388	490		
	CDH74	31909	6254	4917	3916	1790	554	72	83	342	1350	2708	4276	5647		
	CDH80	14830	3293	2467	1711	515	80	3	4	44	464	1121	2128	3000		
(14) Wind	WSAvg	5.5	6.4	6.4	6.0	5.4	5.0	4.7	4.5	4.9	5.2	5.7	5.9	6.1		
(15) Precipitation	PrecAvg	24.4	3.7	3.3	2.2	1.5	1.6	0.9	1.0	1.0	0.9	1.8	2.5	3.9		
	PrecMax	39.0	12.8	10.8	14.2	10.4	7.0	4.1	6.3	6.2	4.0	7.1	6.5	13.4		
	PrecMin	9.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7		
	PrecStd	6.9	2.5	2.5	2.5	1.9	1.6	1.0	1.2	1.3	1.1	1.6	1.5	2.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.2	103.9	98.7	90.7	85.5	78.5	79.3	84.1	96.2	96.6	100.0	103.2		
		MCWB	71.5	69.5	70.2	67.2	64.8	61.5	61.7	60.1	63.4	65.4	68.3	68.1		
	2%	DB	99.3	99.5	93.7	87.5	82.2	76.0	76.2	80.8	89.3	92.2	96.4	99.4		
		MCWB	70.9	69.6	69.9	66.4	62.8	60.6	59.4	60.3	61.5	65.7	67.5	68.7		
	5%	DB	96.6	95.4	91.4	85.1	79.6	73.8	73.8	77.9	85.0	89.2	93.6	96.0		
		MCWB	70.8	70.4	69.7	65.9	62.1	59.1	58.5	58.4	60.5	64.8	66.9	68.6		
	10%	DB	93.6	91.8	88.9	83.0	77.1	71.5	71.3	74.9	81.3	86.3	90.8	93.0		
		MCWB	70.7	70.3	69.6	65.0	61.1	57.6	57.0	56.4	60.0	63.9	67.0	68.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.7	77.3	76.3	71.1	68.4	64.5	66.8	67.0	69.1	71.6	73.8	76.1		
		MCDB	89.6	86.5	83.5	83.0	80.4	72.7	75.0	77.0	75.2	84.3	87.3	87.4		
	2%	WB	76.2	75.8	74.7	69.8	65.9	62.8	64.2	63.7	67.2	70.0	72.0	74.2		
		MCDB	87.4	85.1	82.0	80.4	75.5	72.8	70.8	73.1	74.5	82.5	87.2	84.0		
	5%	WB	75.0	74.5	73.6	68.4	64.1	61.2	62.0	61.4	65.4	68.8	70.6	72.8		
		MCDB	85.9	84.9	82.9	79.1	75.2	70.1	68.9	71.9	75.8	81.4	85.1	83.3		
	10%	WB	73.9	73.4	72.5	67.3	62.7	59.8	59.8	59.2	63.7	67.4	69.3	71.7		
		MCDB	84.3	84.5	82.0	78.1	73.9	68.0	67.1	70.6	76.3	79.8	82.8	83.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.1	19.9	23.7	25.3	24.0	26.5	30.0	28.3	26.0	23.5	21.7		
		MCDBR	25.8	26.2	24.3	26.0	27.3	26.4	28.7	33.0	33.1	29.8	27.8	26.5		
	5% WB	MCWBR	6.3	6.8	6.5	9.4	11.9	12.6	14.4	15.8	12.5	10.0	8.2	6.9		
		MCWBR	19.5	18.8	16.9	21.5	22.6	20.7	19.9	26.0	22.9	22.6	22.5	18.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.380	0.364	0.346	0.329	0.309	0.300	0.291	0.306	0.340	0.369	0.389	0.387		
		taud	2.517	2.575	2.614	2.620	2.614	2.650	2.655	2.585	2.490	2.441	2.430	2.474		
	Ebn at Noon	Ebn at Noon	305	306	303	295	288	285	292	298	300	301	301	304		
		Edh at Noon	36	33	30	28	26	24	24	28	34	38	39	38		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2174	2060	1850	1626	1348	1154	1296	1593	1901	2122	2213	2232		
	RadStd	RadStd	164	178	149	137	70	83	97	97	166	161	244	226		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page